
The Shadow-Software Toll Scam

How cash kept flowing through India's "cashless" highways — and who looked away

CashlessConsumer Research

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How cash kept flowing through India's "cashless" highways — and who looked away

A CashlessConsumer explainer • September 2026 • Version 1.0

In one sentence: At roughly 100 Indian highway toll plazas, operators installed unauthorised “shadow” software on their own booth computers that printed real-looking receipts for cash taken from non-FASTag vehicles — while showing the national toll system almost none of it. The fraud ran for about two years across at least 14 states before a police sting, a parliamentary admission, a botched crackdown, a court reversal and finally a federal money-laundering probe put the spot-light back on it.

Executive summary

What happened. For roughly two years (March 2023 to January 2025), operators at a toll plaza on National Highway 35 in Mirzapur, Uttar Pradesh ran a piece of unauthorised software — known in the case as “Mobdata” and “Any” — on the plaza’s own booth computers. The software let booth operators record payments from vehicles without a valid FASTag sticker, collect cash from those drivers, and then make almost all of those transactions vanish from the official accounting system. Drivers paid. The government did not get its user fee. Only about 5% of the shadow transactions ever entered the official record.

How big. What UP Police’s Special Task Force found at one plaza in January 2025 turned out to be a template. Police and NHAI traced the same pattern to about 42 plazas within weeks; by September 2026, when the Enforcement Directorate filed a money-laundering case, the number was about 100 plazas across more than 14 states, with searches at 14 premises in 6 states. The STF’s working estimate at the original plaza alone was ₹45,000 of skimmed cash per day; police put total diversions across the identified network at over ₹120 crore. NHAI encashed over ₹100 crore of bank guarantees from defaulting agencies in March 2025 — then saw much of that enforcement unwind in the Delhi High Court in July 2026.

Why it matters. This was not a hack. Nobody broke the FASTag encryption or spoofed a tag. The “bypass” was human trust plus unaccountable software on machines that only the operator’s own staff could touch — a hole that every audit report and reconciliation system in the chain was structurally blind to, because the shadow software kept the *appearance* of a normal system: a steady trickle of “exempted” and “violation” receipts, bank remittances that matched the (falsified) books, and drivers walking away with legitimate-looking receipts.

What this report does. It explains the toll-plaza system from scratch for readers who have never thought about what happens at a toll booth; reconstructs the bypass from police chargesheets and press records; measures the footprint against official open data (the IHMCL plaza registry and NHAI’s own Toll Information System); follows the accountability arc through three acts — crackdown, court reversal, and federal money-laundering probe; and critiques the three procurement documents NHAI has published since, which quietly concede that lane-level cash is the system’s unmonitored blind spot.

Why you should read this

If you have ever driven on an Indian national highway, this case is about a system you paid into — and about the small fraction of payments that still happen outside the digital trail. It explains:

1. **How tolling actually works** — the hardware, the software, and the little-known manual side-door that still existed at every plaza.
2. **How the bypass worked** — in plain language, reconstructed from police and enforcement-directorate findings.
3. **Why the audits missed it** — a systems failure worth understanding, because it will happen again in other digital-public-infrastructure systems.

4. **What happened to accountability** — including the part where the High Court undid the regulator's punishment, and what happened next.
5. **What the open data shows** — we matched the scam plazas against the government's own plaza registry, and the results are uncomfortable.
6. **What changes now** — and what risks are simply migrating to the next system.

This report is written for a reader with **zero background** in tolling systems or this case. Every technical claim is sourced; every unverified claim is labelled as such. A full evidence archive underlies this report (see Appendix C).

Part I — The system, before the scam

1. A toll plaza, disassembled: the lane, the reader, the booth computer, and the money

India’s national highways (the “NH” network) are largely funded through **user fee** — the toll you pay at a plaza. There were, as of the government’s own registry from August 2025, **more than 1,100 fee plazas** across 24 states and union territories (we count them in Part IV).

A toll plaza is not run by the government. It is run by a **fee-collection agency** — a private contractor engaged by the concessionaire (the company that built or operates the highway section) under the overall supervision of the **National Highways Authority of India (NHAI)**, a statutory body under the Ministry of Road Transport and Highways (MoRTH). A separate NHAI joint venture, the **Indian Highways Management Company Limited (IHMCL)**, runs the technology backbone of electronic tolling.

At the booth, the stack looks like this:

Layer	What it is	Who controls it
Lane hardware	RFID (FASTag) reader, barrier arm, vehicle-classification sensors, CCTV/ANPR camera, thermal receipt printer	The concessionaire’s vendor
Toll Management System (TMS)	The lane software: reads the tag, classifies the vehicle, computes the fee, records the transaction, prints the receipt	The fee-collection agency’s technology vendor
Back office	Batch settlement of the day’s transactions to the acquiring bank, then to IHMCL/NPCI’s NETC clearing system	Agency + bank
Oversight	NHAI Regional Offices inspect plazas; IHMCL monitors NETC transactions; ministry committees review	Government

The critical sentence in that table is the third row of the first column: **the TMS lane software is procured and controlled by the private collection agency, not by NHAI**. Hold that thought.

2. FASTag: how a windshield sticker came to carry 98 percent of India's highway payments

Since February 2021, paying toll on national highways by cash has been formally the exception, not the rule. Vehicles carry a **FASTag** — an RFID sticker linked to a prepaid wallet or bank account. As you approach the plaza:

- The reader identifies the tag, the lane software computes the fee for your vehicle class, the barrier lifts, your account is debited, and the transaction flows into the **National Electronic Toll Collection (NETC)** ecosystem operated by NPCI (the same organisation behind UPI).
- Radio-frequency tags fail sometimes — dead battery in the reader, damaged sticker, insufficient balance, or a tag listed as blacklisted. For exactly these vehicles, the barrier stays down and the booth operator keys the vehicle class manually, charges a **double fee** (the “2× penalty” for travelling without a valid tag), prints a receipt and accepts **cash**.

In ministerial statements to Parliament, roughly **98% of transactions** were electronic; the non-FASTag residue was about **2% of vehicle volume**¹. Two per cent sounds trivial. It is not. Across a national network handling hundreds of millions of trips, the non-FASTag lane is a constant, cash-heavy, operator-controlled stream — and, as this case shows, it was being **skimmed at industrial scale**.

3. The manual side-door: how vehicles without a valid tag pay — the loophole everything hangs on: how non-FASTag vehicles pay cash - and why that lane was never really watched

Here is the design weakness at the heart of this whole story, and it was visible in official documents years before the scam surfaced:

- The operator's keystroke in the TMS decides what the national system believes happened at the lane. If the operator marks a paying vehicle as “**exempted**” (exempt category) or “**violation**” (jumped the barrier, or could not be charged), no toll is recorded as due.
- There was **no independent physical count** at the lane — no reconciling of barrier events or sensor counts against keyed transactions.
- The agency's own technology vendor controlled the lane software that produced the receipt and the record.
- A program to monitor whether agencies actually **remit** what they collect (“Toll Remittance Alert System”) was only being tendered in April 2023 — *after* the fraud had already begun².

In short: the national system trusted the lane, and the lane belonged to the contractor.

¹Ministerial statements, Lok Sabha, 20-21 Mar 2025; FASTag penetration of “over 98 percent” per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

²IHMCL, “Selection of Agency for Design, Development, Implementation... of Toll Remittance Alert System (TRAS)”, RFP, 3 April 2023 (PDF in evidence archive).

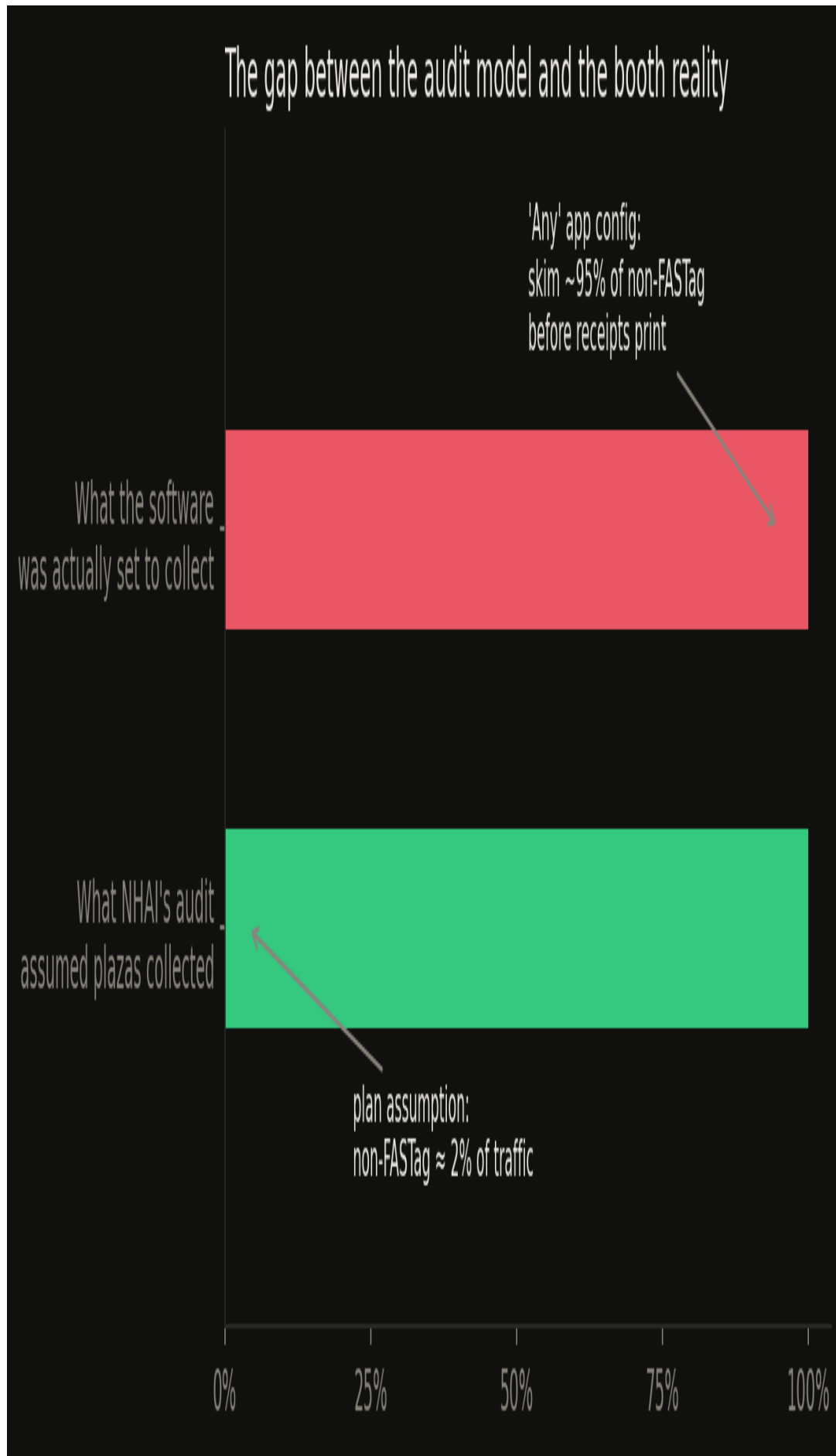


Figure 1: The gap between the audit model and the booth reality. What the official record shows versus what drivers actually paid at a scam plaza: roughly one in twenty shadow transactions entered the TMS, each shadowed by a plausible-looking printed receipt. Chart: CashlessConsumer Research

reconstruction from STF/ED case reporting.

Part II — The scam

4. January 2025: a checkpoint sting at one Mirzapur plaza blows open a network of forty-two

On 22 January 2025, Uttar Pradesh police's Special Task Force (STF) registered FIR No. 0017/2025 at Lal-ganj police station, Mirzapur district, and moved on the **Atraila Shiv Gulam toll plaza** on the Varanasi–Hanumana section (old NH-7, now NH-135) — a plaza on the pilgrimage-and-coal corridor southeast of Mirzapur³.

Within two days the STF had:

- Arrested **Alok Kumar Singh**, 35, an MCA (computer-applications) graduate from Jaunpur district who had briefly worked at a toll plaza in 2021. He was picked up near Babatpur airport, Varanasi⁴.
- Detained the plaza's manager, **Rajeev “Raju” Mishra** of Prayagraj, and an operator, **Manish Mishra** of Sidhi (MP)⁵.
- Seized two laptops, a printer, five mobile phones, ₹19,580 in cash and a vehicle⁶.

What the STF described was not a cash-skimming trick by a rogue operator. It was a **software product**, allegedly built by Singh, that had been **installed on the plaza's own booth computers** and operated in parallel with the official TMS⁷.

5. Anatomy of the bypass: ‘Mobdata’ and ‘Any’, the shadow software that stood between the booth and the books

Pulling together STF briefings, accused interrogations and later ED disclosures, the machine had these parts:

1. The trigger. A vehicle arrives without a valid FASTag. Barrier stays down. Under the rules, the operator must charge double the fee, in cash, and the transaction must be keyed into the official TMS (the 2x rule that took effect 15 Nov 2025)⁸.

2. The shadow app. On the booth computer, alongside the TMS, ran an unauthorised application (referred to in ED statements by the names ‘**Mobdata**’ and ‘**Any**’ — see the caveat in §11). It computed the **same double fee** from the public gazette fee schedules, so the receipt it printed was arithmetically indistinguishable from a genuine one⁹.

³<https://timesofindia.indiatimes.com/city/lucknow/mca-grad-behind-toll-fraud-arrested/articleshow/117498101.cms>

⁴<https://timesofindia.indiatimes.com/city/lucknow/mca-grad-behind-toll-fraud-arrested/articleshow/117498101.cms>

⁵<https://www.aajtak.in/uttar-pradesh/story/toll-plaza-software-scam-fraud-worth-crores-stf-caught-3-accused-varanasi-uttar-pradesh-lclar-strc-2149511-2025-01-23>

⁶OpIndia, 25 Jan 2025: <https://web.archive.org/web/20250720160857/https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

⁷Dainik Bhaskar, 23 Jan 2025. Original link (<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>) is now dead; full archived copy in the public evidence bundle: <https://zo.pub/cashlessconsumer/fastagscam-open-data>

⁸National Highways Fee (Third Amendment) Rules, 2025 - non-FASTag vehicles paying cash charged double fee, effective 15 Nov 2025. PIB, 4 Oct 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2174761>

⁹<https://navbharattimes.indiatimes.com/metro/lucknow/crime/toll-tax-fraud-by-fake-receipts-through-software-at-42-city-of-india-news/articleshow/117503938.cms>

- 3. The forged receipt.** The app used the **same thermal printer path** as the TMS. The driver got a plausible receipt showing the toll and the 2× penalty. Nothing the driver could see was wrong¹⁰.
- 4. The vanish.** Only about **5%** of these non-FASTag transactions were keyed into the official TMS — enough to keep the “exempted/violation” categories looking ordinary. The other 95% existed only inside the shadow app¹¹.
- 5. The monitoring layer.** Plaza-level collections were pushed to a **central monitoring system** — ED describes separate monitoring portals; STF found Singh’s laptop had direct access to plaza data across plazas¹²¹³.
- 6. The split.** Cash stayed at the plaza level; Singh’s share moved through at least five bank accounts and digital wallets, according to ED findings reported in September 2026¹⁴.

Why the scam was invisible from above: how a falsified system still looked perfectly normal

Do the arithmetic the way an auditor would. Non-FASTag traffic is about 2% of volume. Skimming 95% of it removes **under 2% of gross revenue** — a rounding error in aggregate charts, easily masked by genuine monthly variance. And every audit trail NHAI could query — the TMS database, the NETC settlement records, the bank remittances — was fed by the very lane software the shadow app sat beside. **The fraud never wrote to the system that was being audited.**

That is why, when NHAI’s technical team examined computers at plazas in Gorakhpur after the STF raids, it reportedly found “no significant evidence”¹⁵. Auditing the books cannot catch a second set of books.

6. The footprint: from one plaza on NH-35 to about a hundred plazas across fourteen states

Date	Who says	Count
23 Jan 2025	Day-1 interrogation claims	“up to 200 plazas” (confession-based, never confirmed) ¹⁶
23 Jan 2025	STF verified list	42 plazas in 14 states, operators named ¹⁷

¹⁰<https://navbharattimes.indiatimes.com/metro/lucknow/crime/toll-tax-fraud-by-fake-receipts-through-software-at-42-city-of-india-news/articleshow/117503938.cms>

¹¹OpIndia, 25 Jan 2025: <https://web.archive.org/web/20250720160857/https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

¹²OpIndia, 25 Jan 2025: <https://web.archive.org/web/20250720160857/https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

¹³<https://english.mathrubhumi.com/news/crime/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-without-fastag-ed-o0qf4hpy>

¹⁴<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

¹⁵Hindustan Times, Gorakhpur edition, January 2025 (press capture in evidence archive).

Date	Who says	Count
13 Feb 2025	STF	150–200 plazas “under examination”; notices to managers in 6 districts ¹⁸
3 Sep 2026	ED (device forensics)	~100 plazas where the apps were used ¹⁹

¹⁶Dainik Bhaskar, 23 Jan 2025. Original link (<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>) is now dead; full archived copy in the public evidence bundle: <https://zo.pub/cashlessconsumer/fastagscam-open-data>

¹⁷<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

¹⁸<https://www.amarujala.com/uttar-pradesh/mirzapur/notice-to-toll-plaza-managers-of-six-districts-in-120-crore-rupees-scam-in-mirzapur-2025-02-13>

¹⁹The Tribune, 4 Sep 2026: <https://web.archive.org/web/20260904110010/https://www.tribuneindia.com/news/india/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-not-bearing-fastag-stickers-ed/> (live URL intermittently blocks automated access)

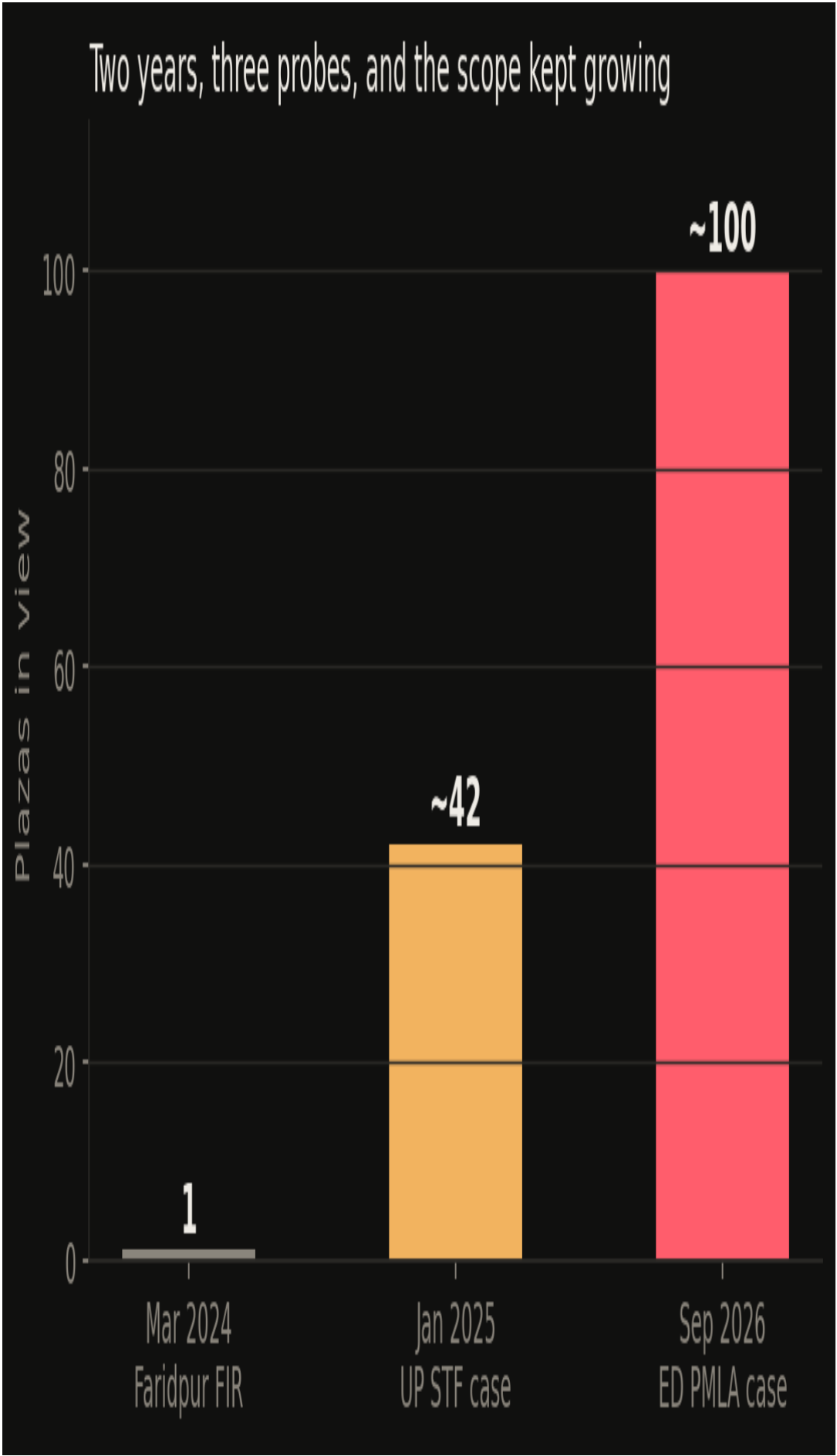


Figure 2: How the number grew. What police found at one plaza in January 2025 was a deployable template; each official acknowledgement expanded the count. The September 2026 ED figure (~ 100 plazas) is an admission count, not a ceiling.

The verified pattern: 42 was the *arrest perimeter*, not the ceiling. ED's forensic examination of seized devices in 2026 nearly tripled the confirmed footprint, and pointed to plazas in **UP, Rajasthan, Madhya Pradesh, Jammu & Kashmir, West Bengal and the Delhi-NCR** among the searched premises²⁰²¹.

²⁰<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

²¹<https://www.dailyexcelsior.com/about-100-highway-plazas-including-some-in-jk-used-illegal-apps/>

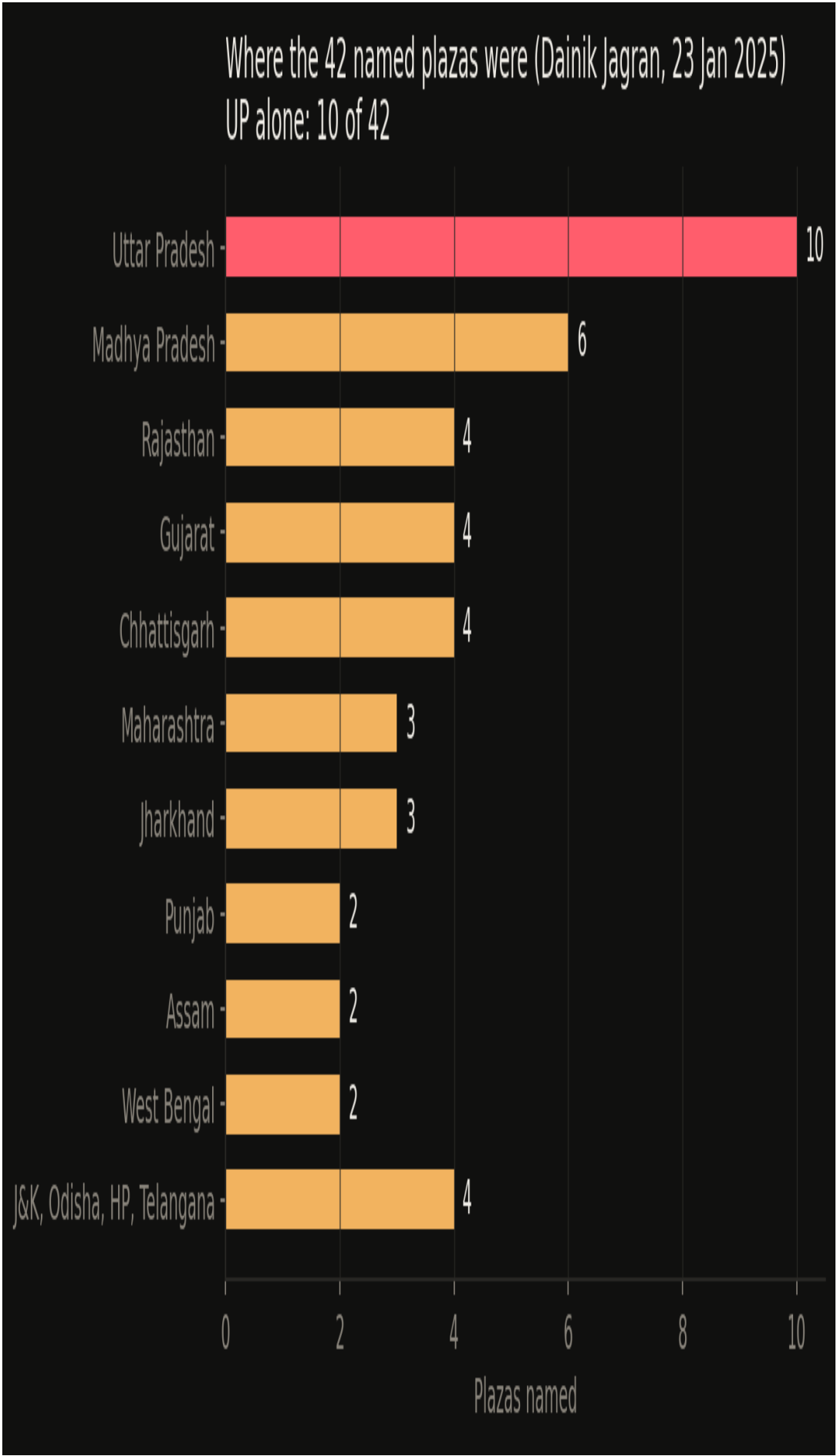


Figure 3: Where the 42 verified plazas were. State-wise split of the STF’s January 2025 verified list, as reported by Dainik Jagran. UP, MP and Rajasthan alone account for 20 of the 42. Source: press release by CashlessConsumer Research, compile in the evidence archive.

7. Following the money: ₹45,000 a day at one plaza, ₹120 crore across the network, ₹100 crore in guarantees

- **Per plaza:** STF's working estimate at Atraila was roughly **₹45,000 per day** of skimmed cash²². Over the ~730 days the fraud reportedly ran, that is about **₹3.3 crore from a single plaza** — a figure Dainik Bhaskar separately reported as the “3.28 crore embezzled at one location”²³.
- **Network scale:** the STF put total diversion at **₹120 crore+** across the 42 plazas²⁴. Early day-1 media claims reached ₹750 crore — never substantiated, treat with caution.
- **ED's 2026 language:** receipts for unauthorised collections “running into crores” across ~100 plazas; ₹1.03 crore cash seized in the September 2026 searches and **eight bank accounts** frozen²⁵.

8. The humans: an MCA graduate, a booth-operator network, and the agencies that hired them: an MCA graduate, a plaza manager, a booth-operator network, and the agencies that hired them

Actor	Role	Status
Alok Kumar Singh	Alleged developer of the shadow software; MCA, ex-toll-plaza worker	Arrested Jan 2025
Rajeev “Raju” Mishra	Manager, Shivgulam plaza (Prayagraj native)	Arrested Jan 2025
Manish Mishra	Booth operator (Sidhi, MP)	Arrested Jan 2025
Saaban Lal Kumhavat	Fourth arrest	Feb 2025 ²⁶
“Sawant” and “Sukhantu”	Alleged system managers for the 42-plaza operation	Named in STF briefing; status unclear ²⁷
“Veltek” (per Dainik Jagran)	Collection agency named for Atraila plaza	Identity unresolved; not among the debarred 14 ²⁸

²²<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

²³Dainik Bhaskar, 23 Jan 2025. Original link (<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>) is now dead; full archived copy in the public evidence bundle: <https://zo.pub/cashlessconsumer/fastagscam-open-data>

²⁴<https://www.thedailyjagran.com/india/toll-collection-fraud-nhai-bans-14-agencies-for-using-fake-software-to-collect-toll-illegally-rs-100-cr-security-confiscated-10225098>

²⁵<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

²⁶<https://www.amarujala.com/uttar-pradesh/mirzapur/notice-to-toll-plaza-managers-of-six-districts-in-120-crore-rupees-scam-in-mirzapur-2025-02-13>

²⁷OpIndia, 25 Jan 2025: <https://web.archive.org/web/20250720160857/https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

²⁸<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

Publishing caution: several names above come from single-outlet or briefing-based reporting. We label them as alleged, and we do not name private individuals beyond what multiple independent reports establish.

Part III — Accountability, in three acts

9. Act one (March 2025): NHAI strikes back - fourteen agencies blacklisted, ₹100 crore of securities encashed

The STF case triggered the fastest nationwide contractual crackdown NHAI had executed:

- In Parliament (20–21 March 2025), Minister Nitin Gadkari confirmed the bypass: vehicles without valid tags were being recorded as “exempted or violation” with receipts never entering the system — while insisting there was **no breach of the electronic tolling system itself**²⁹. Announcements followed: AI-based audit cameras and ANPR pilots.
- NHAI **debarred 14 user-fee collection agencies** for one to two years, **encashed performance securities reported at over ₹100 crore**, and levied penalties exceeding ₹2 crore^{30,31}.
- Among the debarred: AK Construction (linked to nine plazas on the scam list), RK Jain Infra Projects, Vanshika Construction, Innovision Ltd and others³². The Atraila agency was separately terminated with its bank guarantee encashed.

10. Act two (July 2026): the Delhi High Court unwinds the blacklist

In July 2026, the **Delhi High Court (Justice Sachin Datta)** set aside NHAI’s debarments, contract terminations and bank-guarantee encashments³³. The court’s reasoning, as reported:

- **No agency employee was named in any charge sheet** — the prosecuted individuals were the software operator’s network, not the contractors’ staff.
- **No shadow software was recovered from the petitioners’ plazas** in any seizure.
- NHAI’s own inspections had found nothing; the action rested on a press note, a confession attributed to a third party, and **statistical inference** (the revenue pattern) — which the court held was not credible material for such penalties.

This is the quiet earthquake of the case. The entire contractual accountability layer — the ₹100 crore in securities, the debarments, the terminations — **collapsed**. Whatever the ultimate truth about what collection agencies knew, the *evidence* assembled through the contractual route did not survive judicial scrutiny.

²⁹ Ministerial statements, Lok Sabha, 20-21 Mar 2025; FASTag penetration of “over 98 percent” per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

³⁰ <https://money.rediff.com/news/market/nhai-debars-14-toll-agencies-for-irregularities/23868620250321>

³¹ <http://www.uniindia.com/nhai-bans-14-agencies-for-irregular-activities-in-fee-collection-at-toll-plazas/india/news/3418421.html>

³² <https://money.rediff.com/news/market/nhai-debars-14-toll-agencies-for-irregularities/23868620250321>

³³ <https://indialegallive.com/constitutional-law-news/courts-news/delhi-high-court-sets-aside-nhai-action-against-toll-contractors-in-software-fraud-case/>

11. Act three (September 2026): the Enforcement Directorate arrives, and the case becomes about laundering

Which brings us to the news that started this investigation: on **2 September 2026**, the **Enforcement Directorate (ED)** — the federal financial-crime agency — conducted searches at **14 premises across six states** under the Prevention of Money Laundering Act (PMLA), seized ₹1.03 crore in cash and froze eight bank accounts³⁴. On 3 September, the ED disclosed that its forensic examination of devices pointed to the unauthorised apps and monitoring portals at **about 100 plazas**³⁵³⁶.

Read in sequence, the ED action is the state's **second attempt at accountability** — this time using criminal-forensic machinery (device images, banking trails, portal data) rather than contractual penalties, precisely because the contractual route failed in court.

On the names ‘Mobdata’ and ‘Any’: these app names appear only in ED-attributed wire copy. We searched app stores, code-hosting platforms, and certificate-transparency logs; **no public trace of any product by these names exists**. They are likely internal codenames visible on the seized devices and portals. We treat them as attributed claims, not established product names — and we note that “MobData” is also the name of an unrelated Chinese analytics SDK, which should not be conflated.

³⁴<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

³⁵The Tribune, 4 Sep 2026: <https://web.archive.org/web/20260904110010/https://www.tribuneindia.com/news/india/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-not-bearing-fastag-stickers-ed/> (live URL intermittently blocks automated access)

³⁶<https://english.mathrubhumi.com/news/crime/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-without-fastag-ed-o0qf4hpy>

Part IV — What the open data shows

This is the part you cannot get from news reports, so we built it.

12. Counting every plaza: what the official registry says (and lets anyone check)

IHMCL publishes the official list of national-highway fee plazas as a PDF³⁷. We parsed it (methodology in Appendix A) and extracted **1,133 plazas**, each with its NETC plaza code, name, state and district.

Where the plazas are:

State	Plazas
Rajasthan	163
Uttar Pradesh	135
Maharashtra	101
Madhya Pradesh	89
Tamil Nadu	81
Andhra Pradesh	79
Haryana	78
Karnataka	62
Gujarat	56
Punjab	41
Bihar	41
Telangana	36
Odisha	32
West Bengal	29
Chhattisgarh	23
Jharkhand	22
Delhi	13
Assam	10
Kerala	8
Uttarakhand	8
Jammu & Kashmir	7 (+1)
Himachal Pradesh	5

³⁷IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

State	Plazas
Meghalaya	4

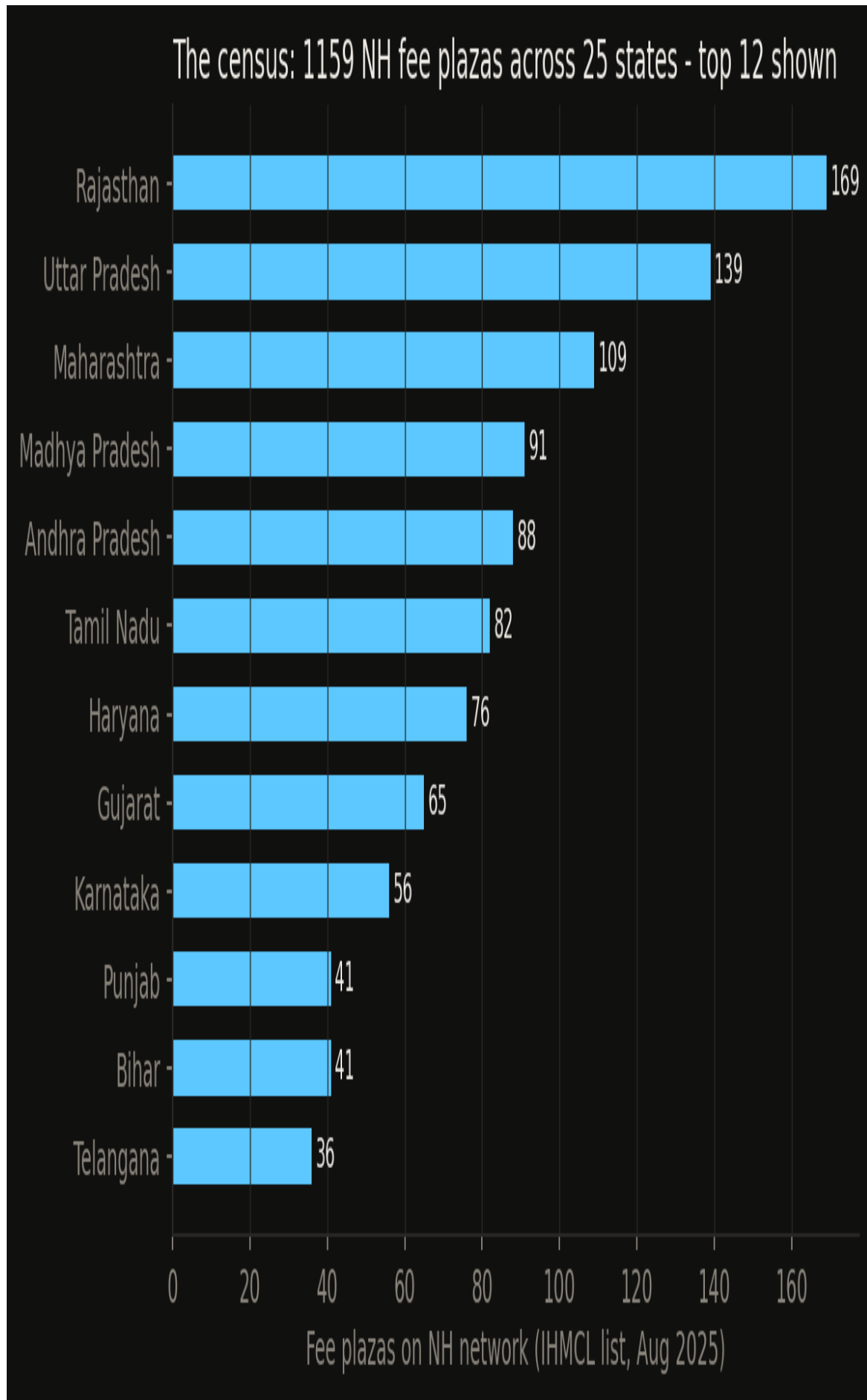


Figure 4: The official registry, counted by state. 1,129 National Highway fee plazas across 22 states/UTs in the IHMCL list of August 2025 – the same registry that still lists Atraila Shiv Gulam today. CashlessConsumer Research

Source: IHMCL NH Fee Plaza list, Aug 2025 (see Appendix D).

(Parsing note: ~1,133 rows parsed cleanly of a reported ~1,150+; a handful of wrapped rows could not be state-resolved and are excluded from state counts. Raw CSV and script are in the evidence archive.)

The headline: the scam plazas were concentrated in exactly the states where plazas are densest — **UP (135 plazas in the registry) contributed the largest single block of the 42-plaza list (10)** — but the footprint is genuinely national: 14 states in the STF list alone, from Jammu to Assam.

The second open-data source: NHAI's own Toll Information System

The IHMCL list is only names and locations. NHAI's **Toll Information System** (TIS, tis.nhai.gov.in) goes further: it is the ministry's own map-based registry, and its underlying data feed carries, per plaza, the fee-notification date, the commercial operation date, the **concessionaire/operator of record**, traffic versus target figures, and per-vehicle fee schedules — the exact fields an auditor would need to reconcile what a plaza *should* have collected against what it *did* remit.

That feed is now heavily access-controlled, but a full snapshot from August 2022 (782 plazas with structured fields) survives in public web archives and is republished with this report³⁸. It matters for three reasons. First, it shows what *was* technically feasible years before the scam surfaced: per-plaza operator identity, fee schedules and coordinates in machine-readable form. Second, it records Atraila Shiv Gulam as a **hybrid-annuity-model (HAM) plaza** with an individual operator of record — the structure that let one contractor's booth computers run unauthorised software for years. Third, its fee-schedule fields let anyone reproduce the arithmetic of a skim: at Atraila's recorded car fee of ₹80, the STF's claimed ₹45,000/day of pocketed cash implies roughly **560 unticketed car-equivalents a day** at a single booth.

³⁸<https://web.archive.org/web/2022/https://tis.nhai.gov.in/TollPlazaService.aspx/GetTollPlazaInfoForMapOnPC>

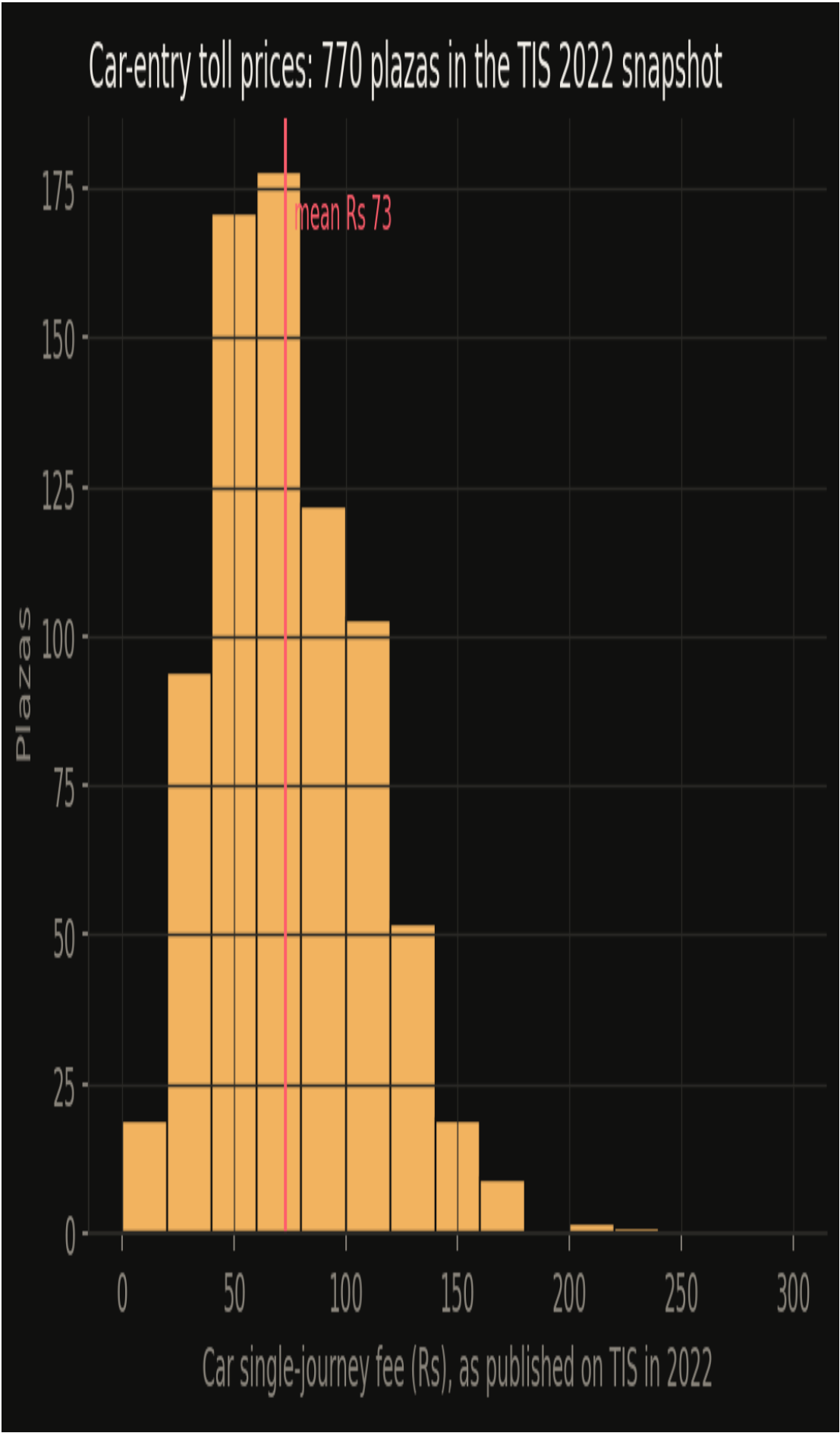


Figure 5: What one car pays, plaza by plaza - distribution of single-journey car fees across 782 plazas in NHAI's August 2022 Toll Information System snapshot. The long right tail (₹300+) is where a cashlessConsumer Research 23 skim earns the most per unticketed vehicle. Source: NHAI TIS snapshot, Aug 2022 (see Appendix D).

The scandal within the open data is the *decay*: the 2022 feed was richer in audit-relevant fields than anything NHAI publishes openly today. Rebuilding that transparency — operator identity, fee notifications, and reconciliation data per plaza — is a cheaper control than any camera.

13. Cross-walking the scam list against the registry: Atraila is still listed: Atraila is still listed

We matched the reported scam plazas against the official registry. Three results matter:

1. The scam plazas are ordinary plazas. Atraila Shiv Gulam appears in the registry as *Atralia SivGulam*, NETC code **320132**, district Mirzapur, on the “Varanasi to Hanumana pkg-III” section — no flag, no annotation, indistinguishable from the 1,132 honest plazas³⁹. The same for Patni Pratappur (NETC 536051, Shamli district) from the scam list. **There is no public marker in the government’s own data that anything was ever wrong at these plazas.**

2. The registry still lists them. As of the August 2025 edition — after the arrests, after the debarments — the plazas remain listed. We take no position on whether they should have been removed; but a citizen or journalist using only official data would never learn that a scandal had touched these locations.

3. The crosswalk is incomplete — by design, perhaps. The Baghpat scam plaza cannot be uniquely matched to one of the five Baghpat-district plazas in the registry from press names alone. Names in press lists (“Bagpat”, “Chikli”, “Shauly”) do not cleanly match registry names (“Jiwana”, “Baleni”, “Bada-gaom”...). **Matchable only via the concession contract or NETC code — neither public.** An RTI for the 42-plaza list with NETC codes is the obvious next step (see §19).

13A. The ToT question: are any scam plazas institutional-leased assets?

Why it matters. Since 2018, NHAI has been leasing mature public-funded plazas to institutional investors under the Toll-Operate-Transfer (ToT) model: 15-30 year concessions in exchange for an upfront premium. Bundle 1 (681 km, ten plazas, Andhra Pradesh + Gujarat Saurashtra) went to Macquarie’s MAIF for Rs 9,681.5 crore in 2018⁴⁰; Bundle 3 (566 km, nine plazas, UP + Jharkhand + Bihar + Tamil Nadu) went to Cube Highways for Rs 5,011 crore in 2020⁴¹; later bundles covered the Allahabad Bypass (Cube, 2023) and the Lalitpur-Sagar stretch (IRB Trust, 2023)⁴². If the shadow-app scam had penetrated a ToT asset, the story would implicate global infrastructure capital - not just low-bid local concessionaires. So: were any of the 42 plazas ToT assets?

³⁹IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

⁴⁰Hindustan Times, 6 Mar 2018: “Australian firm pays govt Rs 9,681 cr to operate nine highway stretches” — <https://www.hindustantimes.com/india-news/australian-firm-pays-govt-rs-9-681-cr-to-operate-nine-highway-stretches/story-kagSI9JZSFqpcUb6YifAul.html>; domain-b, 20 Oct 2020 (Bundle 3: 9 plazas, UP/Jharkhand/Bihar/TN, Cube, Rs 5,011 cr; Bundle 1: 681 km, 10 plazas, MAIF) — <https://www.domain-b.com/economy/infrastructure/roads/nhai-receives-rs5-011-cr-upfront-for-bundle-3-of-toll-highway-projects>

⁴¹Indian Infrastructure, “TOT model shows sustained progress” — bundle history incl. TOT-16 (Telangana NH-44, 2024) and TOT-18 (Odisha, Jan 2026) — <https://indianinfrastructure.com/?p=265905>

⁴²Economic Times, Feb 2023: Cube wins TOT-11 (Allahabad Bypass, Rs 2,156 cr), IRB Trust wins TOT-12 (Lalitpur-Sagar, Rs 4,428 cr) — <https://economictimes.indiatimes.com/industry/transportation/roadways/cube-irb-top-bidders-for-nhai-tot-13-14-auction/articleshow/106079036.cms>

Method. The July 2026 official plaza master carries a concession-type marker for every plaza: TOT, PF (public funded), Conc. (concessionaire), or STATE (state highway authority). We parsed the master and resolved every legible row for the 42 scam plazas.

Result: zero ToT overlap. Every scam plaza that resolves cleanly - Harro, Mungari, Umapur, Baghat, Faridpur, Patni Pratappur, Atraila Shiv Gulam, Nainsar, Mohtara, Shahpura, Mokha, Rohisa, Okha Mandi, Kuchadi, Maharajpur, Kumhari, Bhojpuri, Madanpur, Balachera, Sanwara, Turup - carries the **PF** or **STATE** marker. Atraila is PF/HAM⁴³. Shahdol sits on an MPRDC (state) concession and “Chikli” matches the state-run Chikliya plaza in Ratlam - plazas outside NHAI’s contract framework entirely. Conversely, the master’s TOT-flagged plazas - the Gujarat Saurashtra cluster (Kutch/Morbi/Rajkot/Porbandar), the Andhra Godavari belt, Jhansi, Lalitpur, Rae Bareilly, Ranchi (Pundag), Ghaziabad (Maple Highways) - share no name with the scam list. The Lalitpur flag cross-checks against the real ToT Bundle 12 (Lalitpur-Sagar, IRB, 2023), confirming the markers track actual bundles⁴⁴.

What this means. The scam was not random: it grew in the exact institutional vacuum the registry allows us to see. ToT plazas are few, large, and run under 30-year balance-sheet economics by investors who audit collections daily; a shadow app on a Macquarie or Cube plaza would be found within one reconciliation cycle. The 42 were the opposite - small, low-traffic, two-year user-fee contracts won by minimal bidders, where the operator’s upside was the cash itself. And the state-highway plazas in the list (MPRDC, UP state roads) sit outside NHAI’s TOPL/CMS visibility altogether.

Caveats. Name-matching against a garbled PDF text layer is imperfect: roughly a dozen scam names (Chikhlikala, Dashrakhed, Gahra, Jangabani, Mudiapara, Nawasari, Pashchim Madati, Tand Balidha, and others) could not be resolved to a legible master row, so the claim is “no evidence of ToT overlap in every row we could read”, not a proof of full exclusion. The NHAI InvIT’s seed portfolio has not been name-checked against the scam list - filed as an RTI question. The full crosswalk table is in the open-data bundle⁴⁵.

13A. The ToT question: are any of the scam plazas institutional-leased assets?

NHAI has been selling 15-to-30-year leases on finished, public-funded toll roads through the toll-operate-transfer (ToT) programme — Macquarie-AIF paid ₹9,681 crore for the first bundle (681 km, ten plazas in Andhra Pradesh and Gujarat, 2018); Cube Highways paid ₹5,011 crore for the third (566 km, nine plazas in UP, Jharkhand, Bihar and Tamil Nadu, 2019-20); later bundles went to IRB (Lalitpur-Sagar-Lakhnadon, ₹4,428 crore), Cube again (Allahabad Bypass, ₹2,156 crore), and others,

⁴³ NHAI TIS map service, Aug 2022 snapshot (this report’s open-data bundle): Atraila Shiv Ghulam, project type HAM.

⁴⁴ Economic Times, Feb 2023: Cube wins TOT-11 (Allahabad Bypass, Rs 2,156 cr), IRB Trust wins TOT-12 (Lalitpur-Sagar, Rs 4,428 cr) — <https://economictimes.indiatimes.com/industry/transportation/roadways/cube-irb-top-bidders-for-nhai-tot-13-14-auction/articleshow/106079036.cms>

⁴⁵ fastagscam-open-data bundle (zo.pub/cashlessconsumer/fastagscam-open-data): crosswalk CSV + Plaza_Master_July_18_08_2026 source PDF.

with ToT awards continuing into 2026⁴⁶. If a plaza is a ToT asset, its tolling is run by an institution that paid the exchequer upfront for every rupee of fee — an obvious first suspicion for anyone asking where shadow software could hide.

The July 2026 fee-plaza master answers this directly, because it carries a concession-type marker per plaza (TOT / PF / Conc. / STATE). Cross-walking all 42 Jagran-list plazas against the master: **every plaza whose row is legible resolves to PF (public-funded) or a state-highway authority — zero resolve to TOT**. Atraila Shiv Ghulam: PF (its TIS record independently shows project type HAM). Mungari, Harro and Umapur near Prayagraj: PF — they are *not* the Cube-leased Allahabad Bypass. The master's ToT-flagged plazas — the Saurashtra cluster (Morbi/Rajkot/Kutch/Porbandar), the Godavari belt, Jhansi-Lalitpur-Rae Bareli (matching ToT bundles 11-12), Ranchi, Ghaziabad (Maple Highways), Bihar — share no name with the scam list. Two flagged scam items are *state* roads (MPRDC's Shahdol; Chikliya on a Ratlam state highway), which sit outside NHAI's oversight systems entirely.

Why this matters. The scam concentrated exactly where institutional money is absent: public-funded plazas farmed to low-bid user-fee contractors under two-year cancelable contracts, plus state-highway plazas with no NETC-grade audit at all. ToT concessionaires answer to global credit committees and a 30-year P&L; the shadow app ran where nobody had multi-decade skin in the game. That asymmetry is itself a policy finding: monetisation buys the exchequer upfront and creates one tier of tolling with real oversight, while the un-monetised majority runs on the honour system.

Caveats: a dozen list names could not be resolved in the garbled text layer of the master; matching is by name; and the NHAI InvIT's seed portfolio has not been cross-checked here (an RTI question, drafted). The finding is “no evidence of ToT overlap”, not proof of structural exclusion.

14. The companion scams nobody counted: fake receipts and the Baghpat complaint: receipt tampering, pass misuse, and a driver's complaint at Baghpat

The registry also lets you see what *didn't* get counted. In 2025–26, GST-seat reporting and Hindi press documented a parallel SOP at UP plazas including Itaunja and Nawabganj (Lucknow): trucks running on **~900 fake vehicle-registration numbers per month** to dodge toll entirely — a different exploit of the same manual lane⁴⁷. Consumer complaints continued too: in March 2026, a Baghpat commuter produced a receipt **timestamped 8:46 PM for an 11:30 AM crossing** — the shadow-receipt signature persisting in the wild after the “fixes”⁴⁸. That plaza, ironically, is on the debarred-operator's list.

The lesson for DPI watchers: every digital-public-infrastructure system that delegates the *edge* — the booth, the merchant, the CSC, the banking correspondent — to a private contractor inherits this exact

⁴⁶ToT programme context and bundle values: domain-b (Bundle 3 award, 20 Oct 2020): <https://www.domain-b.com/economy/infrastructure/roads/nhai-receives-rs5-011-cr-upfront-for-bundle-3-of-toll-highway-projects> · Hindustan Times (Bundle 1 payment, 2018): <https://www.hindustantimes.com/india-news/australian-firm-pays-govt-rs-9-681-cr-to-operate-nine-highway-stretches/story-kagSI9JZSFqpcUb6YifAul.html> · Economic Times (Tot 11/12 awards): <https://economictimes.indiatimes.com/industry/transportation/roadways/cube-irb-top-bidders-for-nhai-tot-13-14-auction/articleshow/106079036.cms> · Indian Infrastructure (ToT status to Jan 2026): <https://indianinfrastructure.com?p=265905>

⁴⁷<https://upstox.com/news/business-news/latest-updates/how-govt-plans-to-plug-highway-toll-collection-loopholes-after-up-scam-exposure/article-154498/>

⁴⁸<https://www.tv9hindi.com/state/uttar-pradesh/cctv-camera-found-in-women-restroom-baghpat-toll-plaza-misdeeds-including-allegations-of-illegal-extortion-3708444.html>

risk: the edge can run unauthorised software, and the centre's audits will read only what the edge chooses to write.

Part V — The state's stake and the driver's data

15. Why the lane stayed dark: the state was buying data, not control

The exception lane was not an accident that no one priced. It is the residue of a deliberate design choice, made in 2019, about what the toll system was *for*.

When India chose a centralised RFID clearing house over anonymous stored-value instruments (the metro-card model), the argument was never only about queue times. The design — as analysed at the time in *The Wire*⁴⁹ — had three properties worth restating now:

- **A 4 percent digitisation levy on every rupee of toll** — split between issuer bank (₹1.50), acquirer (₹1.25), IHMCL (₹1.00) and NPCI (₹0.25) per ₹100 collected — a recurring fee that contractors recoup from users inside the gazetted rate. The state privatised the upside of collection; the shadow app was the operators' attempt to privatise the cash floor back.
- **Toll receipts as financial collateral.** The same month the FASTag mandate was pushed through, the CCEA cleared toll securitisation and the first toll-operate-transfer (ToT) bundles went to Macquarie (₹9,681 crore) and Cube Highways (₹5,011 crore). The machine-readable toll stream exists substantially so that 15,000+ km of roads can be repriced, bundled and sold against it. The FASTag data is the asset's prospectus.
- **The data itself was the point.** The article's warning — “the ultimate aim... to capture all activities on India's national highways in machine-readable format”, with CCTNS and e-Way Bill integration proposed — reads differently in 2026. What it predicted was not surveillance creep but *audit asymmetry*: enormous institutional investment in what the data can earn, and none in whether the lane event, the keyed record and the remittance are the same thing.

That asymmetry explains the scam's two-year run better than any conspiracy. The data consumer was an investor pricing a ToT bid off NETC flows; nobody's model priced the 8-10 percent of transactions on the booth floor. When NHAI later moved to make FASTag data a monopoly asset and even exempt pieces of the ecosystem from transparency obligations⁵⁰, it confirmed the hierarchy: the aggregate mattered, the receipt did not. The ED's money-laundering case is, in effect, the state finally pricing the receipt.

16. The privacy dimension: what the system knows — and what this scandal will be used to justify

The same 2019 design analysis set out what a centralised toll system means for the person behind the windshield⁵¹; a companion piece that month noted FASTag's straightforward use for movement

⁴⁹Srikanth Lakshmanan, “FASTag: Will Datafication of India's Tolls Boost Highway Development?”, *The Wire*, 14 December 2019: <https://thewire.in/economy/fastag-will-datafication-of-indias-tolls-boost-highway-development> (full archived text in the evidence bundle).

⁵⁰Srikanth Lakshmanan, “FASTag: Will Datafication of India's Tolls Boost Highway Development?”, *The Wire*, 14 December 2019: <https://thewire.in/economy/fastag-will-datafication-of-indias-tolls-boost-highway-development> (full archived text in the evidence bundle).

⁵¹Srikanth Lakshmanan, “FASTag: Will Datafication of India's Tolls Boost Highway Development?”, *The Wire*, 14 December 2019: <https://thewire.in/economy/fastag-will-datafication-of-indias-tolls-boost-highway-development> (full archived text in the evidence bundle).

tracking in the post-26/11 surveillance architecture⁵². Six years on, the points stand — and the scam’s aftermath sharpens them:

- **The centralised database exists.** Every NETC transaction flows through a clearing house that knows vehicle, issuer bank, plaza, timestamp, and amount — one of the largest vehicle-movement databases in the country, built before India had a data protection law (the DPDP Act arrived only in 2023, and toll systems were never retrofitted with a data-governance review).
- **Anonymity was available and declined.** Anonymous prepaid tags — no bank linkage, no traceability — were technically feasible when the system was designed. The mandate instead required KYC’d, bank-linked tags. The TIS fee schedules even publish *local discounts* (district-registered commercial vehicles pay concessionary monthly pass rates): proof that the system’s economics already encode where you live and what you drive.
- **The blacklist is a geofence waiting for a use case.** The clearing house can update tag blacklists within 15 minutes. The stated use is commercial-vehicle enforcement; the capability is disabling a vehicle’s road access nationally at short notice. In 2019 this was a warning; in 2025, NHAI ordered issuer banks to validate registration numbers and blacklist mismatches — a genuine anti-fraud measure for MLFF, and simultaneously a working prototype for mobility control.
- **The scam is now the justification.** Every post-scam fix — ANPR at every lane, MLFF plate-reading, unified ATMS software, VRN validation — deepens data capture. None of the post-scam procurements we reviewed (§18) contains a consumer data-governance clause: retention limits, purpose limitation, access logs, or an independent audit of who queries the movement database. The fraud is real, and so is the risk that it becomes the standing excuse for a surveillance infrastructure nobody chose.

The consumer-position summary is simple: **you were double-charged in a lane that was never audited, by a system that knows exactly where and when, and the data trail that proves your case is owned by institutions that are not accountable to you.** NPCI, the clearing house, has been explicitly structured outside RTI reach⁵³. An RTI to NHAI gets you a debarment order; an RTI to NPCI gets you nothing.

Part VI — The fix, and whether it fixes

17. What changed after January 2025 — and what did not - and what, two years on, still has not

⁵²“After 26/11 Mumbai Attacks, India Has Weaponised Economic Transactions for Mass Surveillance”, The Wire: <https://thewire.in/government/26-11-mumbai-attacks-india-weaponise-economic-transactions-mass-surveillance>

⁵³Srikanth Lakshmanan, “FASTag: Will Datafication of India’s Tolls Boost Highway Development?”, The Wire, 14 December 2019: <https://thewire.in/economy/fastag-will-datafication-of-indias-tolls-boost-highway-development> (full archived text in the evidence bundle).

Control	When	What it does	Verdict
AI audit cameras + ANPR pilots	announced Mar 2025 ⁵⁴	Independent vehicle counting at lanes	The <i>right</i> control (independent count); deployment status unknown
Toll Remittance Alert System (TRAS)	tendered Apr 2023 ⁵⁵	Alerts when agency remittances lag collections	Blind to this fraud class by design — it monitors <i>remitted</i> money; skimmed cash was never booked
Cash ban: FASTag/UPI-only lanes	10–11 Apr 2026	Ends the cash vector entirely	Kills the vector prospectively; transfers friction to consumers without tags
Unified ATMS software (RFP May 2026)	2026	One national plaza-software stack; spec contains no manual-lane concept at all ⁵⁶	Digitises the exception category itself; removes operator discretion
MLFF (barrier-free tolling) acquirer-bank model	pilot RFP Jan 2025 ⁵⁷	No booth, no operator, no cash — ANPR-based charging	Structural fix; per-plaza procurement model will scale slowly
STQC certification scheme for TMS (tamper-evident databases, audit trails)	scheme live ⁵⁸	Certifies the lane software itself	The control that would have <i>detected</i> the shadow app; unclear it was ever procured for plaza TMS

18. The procurement critique: three RFPs that show NHAI knew where the hole was: three official documents that show NHAI knew where the hole was

Three tenders, read side by side, reveal the pattern:

⁵⁴Ministerial statements, Lok Sabha, 20-21 Mar 2025; FASTag penetration of “over 98 percent” per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

⁵⁵IHMCL, “Selection of Agency for Design, Development, Implementation... of Toll Remittance Alert System (TRAS)”, RFP, 3 April 2023 (PDF in evidence archive).

⁵⁶IHMCL/NHAI, “Unified NHAI ATMS Software” RFP Vol-2, 26 May 2026, 239 pp (PDF in evidence archive).

⁵⁷IHMCL, “Selection of Acquirer Bank for FASTag-ANPR based Multi Lane Free Flow (MLFF) User Fee Collection at Nemili”, RFP, January 2025 (PDF in evidence archive).

⁵⁸<https://www.stqc.gov.in/en/toll-management-system-products-certification-scheme>

- **TRAS (2023)** monitors remittances only — its data feeds are the agency’s own declarations. It is a watchdog fed by the watched.
- **Unified ATMS (2026)** is a 239-page spec with state-of-the-art cloud architecture — and **zero occurrences** of “manual lane”, “cash lane” or “handheld device”. The exception category has been designed away rather than instrumented. Good for fraud, but the residual risk migrates to ANPR mis-reads (fake and duplicate plates — already a documented racket⁵⁹) and to digital-exception fraud.
- **MLFF (2025)** is being procured **one plaza at a time** (“selection of acquirer bank... at Nemili”) — re-procurement as a scaling strategy is how you get a patchwork of vendors and a decade of transition.

And the deepest gap: **no signed, hash-chained, lane-event data integrity requirement** appears in any plaza-software procurement we reviewed. The TMS remains software the contractor buys from whomever it likes, certified (if at all) against no public register. STQC’s certification scheme — tamper-proofing, audit trails — exists on paper; we could not find a public record of a single plaza TMS certified under it. *(RTI filed to confirm.)*

19. What we still don’t know - and are filing right-to-information requests to find out

Our RTI applications (drafted, ready to file at rtionline.gov.in⁶⁰) target the remaining unknowns:

1. **FIR 0017/2025 with seizure lists** (SP, Mirzapur) — the primary evidence of what the software actually did.
2. **The 14-agency debarment file**: plaza-wise map with NETC codes, PBG ledger, penalty orders (NHAI) — the accountability ledger the HC gutted.
3. **TRAS status** (IHMCL): vendor, go-live date, alerts generated to date.
4. **The Atraila concession chain** (NHAI RO Varanasi): who “Veltek” legally is, who the concessionaire was.
5. **STQC certification register** for plaza TMS products.

⁵⁹<https://upstox.com/news/business-news/latest-updates/how-govt-plans-to-plug-highway-toll-collection-loopholes-after-up-scam-exposure/article-154498/>

⁶⁰Workspace archive: Documents/research/FASTagScam/rti/RTI-DRAFTS.md.

Part VII — What it means

20. What it means for you: the driver who pays and the driver who pays twice

If you travelled a non-FASTag vehicle through any of the 42 (now ~100) plazas between roughly 2023 and 2025, you probably **paid double toll in cash and received a receipt that never entered any government system**. Your legal remedy as an individual is effectively nil — the amount is small, the counterparty anonymous, and refunds for all faulty collections nationwide in 2024 totalled ₹12.55 lakh⁶¹ — against a diversion measured in the hundreds of crores. The real loser is the exchequer; the real payer is you again, through taxes and future toll resets.

What you *can* do:

- **Keep FASTag active and linked** — the manual lane is the risk surface; digital lanes are settled and traceable.
- If you get a suspicious toll receipt (wrong timestamps, wrong plaza, amounts that don't match the schedule), report it to NHAI's helpline (1033) **and** to us — the Baghpat complaint shows the pattern persists.
- Support the boring fixes: independent lane counting, public concession data, certified lane software. That is where this stops.

21. What it means for digital public infrastructure: the blind spot is a design pattern

This case is a template, and it should be studied beyond highways:

- **The edge is the attack surface.** UPI's QR codes, ONDC's seller apps, AgriStack's field agents — every DPI stack delegates its edge. An unauthorised app at the edge, showing the centre what the centre expects to see, is a *general* attack pattern.
- **Auditing the ledger is not auditing the world.** NHAI audited its TMS. The TMS was fed by the contractor's lane software. The shadow app never touched the TMS. **Reconciliation must anchor to physical events** (vehicle counts, not keyed records).
- **Punishment is not accountability.** NHAI's ₹100 crore crackdown evaporated in court because the evidentiary work hadn't been done. The ED probe is attempt two. In DPI governance, the sequence matters: forensics first, penalties second.
- **Deleting the vector beats policing it.** The April 2026 cash ban and the manual-lane-free ATMS spec do more than any audit ever did. The honest framing of India's cashless push is that **cash-outsided exceptions were the fraud surface all along** — but the honest cost accounting must also note that each deleted exception creates a new dispute class (plate spoofing, digital-notice fraud) that consumer bodies must now watch.

⁶¹Ministerial statements, Lok Sabha, 20-21 Mar 2025; FASTag penetration of “over 98 percent” per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

Appendix A — Methodology: the registry parse

- **Source:** IHMCL “National Highway (NH) Fee Plazas” list, PDF, August 2025 edition⁶².
- **Extraction:** `pdftotext -layout`, then a Python parser over the row grammar `S . No | NETC code | Name | State | District | Section | NH | PIU | RO`, with next-line fallback for wrapped name cells and a special case for the Atraila row (“Atralia SivGulam”).
- **Result:** 1,133 rows parsed with state attribution; ~9 unresolved (wrapped rows). NETC codes verified by spot-check against known plazas (e.g., 320132 = Atralia SivGulam; 536051 = Patni Pratappur).
- **Crosswalk:** scam-plaza names from the Dainik Jagran 42-plaza list⁶³ were matched on normalised name + district. Unique matches: Atraila Shiv Gulam (320132), Patni Pratappur (536051), Madanpur (Assam). Baghpat and several others are ambiguous (multiple registry plazas per district); we do not force matches.
- **Limitations:** the registry names plazas by their fee-schedule names; press lists use local names. Concessionaire/agency attribution per plaza is **not** in the public registry — it requires RTI or concession documents.
- All scripts, the parsed CSV, and the source PDF are in the evidence archive (Appendix C).

Appendix B — The 42-plaza list (preserved)

From Dainik Jagran, 23 January 2025⁶⁴ — preserved here because the original article is already dead-linking:

Uttar Pradesh (10): Harro (Prayagraj) · Mungari (Prayagraj) · Umapur (Prayagraj) · Andi Lohra (Azamgarh, AK CC) · Bagpat (AK CC) · Faridpur (Bareilly) · Patni Pratappur (Shamli) · Atraila Shiv Gulam (Mirzapur) · Nainsar (Gorakhpur) · Chikli **Madhya Pradesh (6):** Jangabani · Mohtara (AK CC) · Shalibada · Shahdol · Gahra · Chikhlikala (Chhindwara) **Rajasthan (4):** Phulera (Jaipur) · Kadi Shehna (AK CC) · Shahpura · Shaully (operating company reported as “Anuvejan”, ≈ Innovision — debarred) **Gujarat (4):** Mokha (AK CC) · Rohisa (AK CC) · Okha Mandi · Kuchadi **Chhattisgarh (4):** Maharajpur · Mudiapara · Kumhari (Durg, AK CC) · Bhojpuri **Maharashtra (3):** Dashrakhed · Khani Bade · (third per list) **Jharkhand (3):** Nawasari · Turup (AK CC) · Tand Balidha **Punjab (2):** Dhulal · Jigha (AK CC) **Assam (2):** Madanpur (RK Jain Infra — debarred) · Balachera **West Bengal (2):** Gobari · Pashchim Madati **Jammu (1):** Van · **Odisha (1):** Kadali Garh · **Himachal (1):** Sanwara · **Telangana (1):** Jangaon

AK CC = AK Construction (debarred). Counts per Jagran; the outlet’s state tallies differ slightly from other outlets (we preserve the primary list as published).

⁶²IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

⁶³<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

⁶⁴<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

Appendix C — Evidence archive and negative results

This report is the analysis layer; the archive is the evidence layer. All 23+ press captures (with canonical URLs), five government PDFs (TRAS RFP, Unified ATMS RFP, MLFF RFP, plaza registry, ATMS training deck), the parsed registry CSV, RTI drafts and the master dossier live in the workspace under Documents/research/FASTagScam/.

What we looked for and did not find (as of 4 September 2026) — published for scientific honesty:

- No app-store, GitHub, code-hosting or certificate-transparency trace of ‘Mobdata’/‘Any’ or plausible variants.
- No public copy of the ED press release (site’s Umbraco API confirms 1,995 releases through 2 September 2026; none on this case yet).
- No public copy of FIR 0017/2025 or the charge sheet (RTI route).
- No public record of any plaza TMS certified under STQC’s scheme.
- The Lok Sabha answer PDF (USQ 1017) was unreachable from our network; the ministry’s reply is quoted via secondary reporting⁶⁵.
- Caution: “MobData” (MobTech, Chinese analytics SDK) is an unrelated namesake. Several LinkedIn profiles named “Alok Kumar Singh” are unrelated persons; do not chase.

Appendix D — Sources

ED/PMLA (Sep 2026): The Tribune⁶⁶ · Daily Excelsior⁶⁷ · Times of India⁶⁸ · Mathrubhumi⁶⁹ · Bizz Buzz⁷⁰

STF case (Jan–Feb 2025): Times of India⁷¹ · Aaj Tak⁷² · Dainik Bhaskar⁷³ · Dainik Jagran^{74,75} · Navbharat Times⁷⁶ · Amar Ujala⁷⁷ · OpIndia⁷⁸ · Hindustan Times (Gorakhpur)⁷⁹

Official response (Mar 2025): Lok

⁶⁵Ministerial statements, Lok Sabha, 20-21 Mar 2025; FASTag penetration of “over 98 percent” per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

⁶⁶The Tribune, 4 Sep 2026: <https://web.archive.org/web/20260904110010/https://www.tribuneindia.com/news/india/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-not-bearing-fastag-stickers-ed/> (live URL intermittently blocks automated access)

⁶⁷<https://www.dailyexcelsior.com/about-100-highway-plazas-including-some-in-jk-used-illegal-apps/>

⁶⁸<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

⁶⁹<https://english.mathrubhumi.com/news/crime/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-without-fastag-ed-o0qf4hpy>

⁷⁰<https://www.bizzbuzz.news/crime/ed-raids-14-locations-across-india-in-toll-plaza-scam-seizes-over-rs-1-crore-cash-1403756>

⁷¹<https://timesofindia.indiatimes.com/city/lucknow/mca-grad-behind-toll-fraud-arrested/articleshow/117498101.cms>

⁷²<https://www.aajtak.in/uttar-pradesh/story/toll-plaza-software-scam-fraud-worth-crores-stf-caught-3-accused-varanasi-uttar-pradesh-lclar-strc-2149511-2025-01-23>

⁷³Dainik Bhaskar, 23 Jan 2025. Original link (<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>) is now dead; full archived copy in the public evidence bundle: <https://zo.pub/cashlessconsumer/fastagscam-open-data>

⁷⁴<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

⁷⁵<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

⁷⁶<https://navbharattimes.indiatimes.com/metro/lucknow/crime/toll-tax-fraud-by-fake-receipts-through-software-at-42-city-of-india-news/articleshow/117503938.cms>

⁷⁷<https://www.amarujala.com/uttar-pradesh/mirzapur/notice-to-toll-plaza-managers-of-six-districts-in-120-crore-rupees-scam-in-mirzapur-2025-02-13>

⁷⁸OpIndia, 25 Jan 2025: <https://web.archive.org/web/20250720160857/https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

⁷⁹Hindustan Times, Gorakhpur edition, January 2025 (press capture in evidence archive).

Sabha written answers⁸⁰ · Rediff/PTI⁸¹ · UNI⁸² · Daily Jagran⁸³ · Upstox/PTI analysis⁸⁴ **Legal:** India Legal Live (Delhi HC, 6 Jul 2026)⁸⁵ **Policy & tech documents:** IHMCL TRAS RFP⁸⁶ · Unified ATMS RFP⁸⁷ · MLFF Nemili RFP⁸⁸ · IHMCL plaza registry⁸⁹ · STQC TMS certification scheme⁹⁰ **Consumer corroboration:** TV9 Hindi (Baghpat, 5 Mar 2026)⁹¹ **RTI drafts:** workspace archive⁹²

Report by CashlessConsumer research (agentic workflow, human-supervised). Feedback: cashlessconsumer@zo.computer. This report will be updated when the ED press release and RTI responses land. Companion page: cyber.cashlessconsumer.in.

⁸⁰Ministerial statements, Lok Sabha, 20-21 Mar 2025; FASTag penetration of “over 98 percent” per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

⁸¹<https://money.rediff.com/news/market/nhai-debars-14-toll-agencies-for-irregularities/23868620250321>

⁸²<http://www.uniindia.com/nhai-bans-14-agencies-for-irregular-activities-in-fee-collection-at-toll-plazas/india/news/3418421.html>

⁸³<https://www.thedailyjagran.com/india/toll-collection-fraud-nhai-bans-14-agencies-for-using-fake-software-to-collect-toll-illegally-rs-100-cr-security-confiscated-10225098>

⁸⁴<https://upstox.com/news/business-news/latest-updates/how-govt-plans-to-plug-highway-toll-collection-loop-holes-after-up-scams-exposure/article-154498/>

⁸⁵<https://indialegallive.com/constitutional-law-news/courts-news/delhi-high-court-sets-aside-nhai-action-against-toll-contractors-in-software-fraud-case/>

⁸⁶IHMCL, “Selection of Agency for Design, Development, Implementation... of Toll Remittance Alert System (TRAS)”, RFP, 3 April 2023 (PDF in evidence archive).

⁸⁷IHMCL/NHAI, “Unified NHAI ATMS Software” RFP Vol-2, 26 May 2026, 239 pp (PDF in evidence archive).

⁸⁸IHMCL, “Selection of Acquirer Bank for FASTag-ANPR based Multi Lane Free Flow (MLFF) User Fee Collection at Nemili”, RFP, January 2025 (PDF in evidence archive).

⁸⁹IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

⁹⁰<https://www.stqc.gov.in/en/toll-management-system-products-certification-scheme>

⁹¹<https://www.tv9hindi.com/state/uttar-pradesh/cctv-camera-found-in-women-restroom-baghpat-toll-plaza-misdeeds-including-allegations-of-illegal-extortion-3708444.html>

⁹²Workspace archive: Documents/research/FASTagScam/rti/RTI-DRAFTS.md.

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